

Early boreal forest understory plant community development in reclaimed oil sands

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ABSTRACT

The understory is an important reservoir of plant biodiversity and a driver of ecological processes in Canada's boreal forest, making its re-establishment following reclamation of oil sands mines crucial. This study examined understory plant community development with time using two cover soil types (forest floor mineral mix (FMM) and peat mineral mix (PMM)) in Alberta oil sands reclamation sites. Cover soil type played a significant role in plant community development with FMM soil having greater species richness, diversity and total vegetation cover than PMM four years after reclamation. Nonmetric multi dimensional scaling and multi response permutation procedure revealed no compositional differences between the cover soil types. Indicator species analysis showed FMM cover soil was dominated by perennial species while PMM was dominated by annual forb species. This implies that FMM cover soil leads to faster vegetation recovery (from ruderal and annual communities to perennial communities) than PMM cover soil. Non-native species declined with time in FMM (7.5–4%) and increased in PMM (8.9–12.5%). We therefore recommend utilizing FMM as cover soil to promote productive and diverse understory plant communities on the reclaimed landscape.

1. Introduction

The advancement of science and technology, economic development, industrial expansion and human population growth contributes greatly to global forest ecosystem degradation (Aerts and Honnay, 2011). Restoration of forest biodiversity, community structure and associated ecological complexity is a key task in contemporary ecology (Hobbs and Harris, 2001; Palmer et al., 1997) to provide a sustainable ecosystem that is resilient to perturbation (Urbanska et al., 1997; SER, 2004). It is not always obvious when we have reached such conditions. Some researchers suggest restoration success could be based on vegetation characteristics (Wilkins et al., 2003) while others focus on species diversity and abundance or ecosystem processes (van Aarde et al., 1996; Rhoades et al., 1998; Reay and Norton, 2002). In both cases, understory vegetation development in degraded land may be a potential important determinant of restoration success (Koch, 2007; Macdonald et al., 2015a; Dhar et al., 2018, 2019, 2020a).

Restoration strategies depend on the nature of ecosystem degradation or land use history (Hobbs and Harris, 2001; Dhar et al., 2020b) and the outcomes of reclamation often depend on multiple factors including abiotic site conditions, biotic factors and time since reclamation (Stubble et al., 2017; Dhar et al., 2020c). Abiotic site conditions such as

cover soil types can play significant roles in plant community development, whereas biotic factors such as microbial activity help soil formation and nutrient cycling. Cover soils such as peat and forest floor material typically contain seeds and vegetative propagules that can be beneficial for establishing the plant community during reclamation (Gonzalez-Alday et al., 2011). The seed reserve in cover soil is a source of inexpensive, diverse and ecologically adapted vegetation (MacKenzie and Naeth, 2010; Naeth et al., 2013) and may be the only reserve of propagules from which some species can establish (Koch and Ward, 1994; Ward et al., 1996). According to Ward et al. (1996) more than 70% of the species found on a reclaimed site originate from propagules contained in the salvaged cover soil. The use of salvaged cover soil can accelerate reclamation of soil structure and function (organic matter content, nutrient concentrations, microorganisms, water holding capacity, etc.) (Sydnor and Redente, 2002); thus salvaged cover soils are widely used in mine reclamation across the planet. For example, forest floor soil used for bauxite (Ward et al., 1996) and sand (Rokich et al., 2000) mine reclamation in Australia; coal mine reclamation in Chile (Valenzuela et al., 2018), Germany (Ulrich et al., 2016), Mexico (Douterlungne et al., 2018) and Spain (Gonzalez-Alday et al., 2011); granite quarry reclamation in Hong Kong (Zhang et al., 2013) and oil sands mines in Canada (Naeth et al., 2013). Therefore, sites should be

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Table 1
Characteristics of the oil sands five reclamation sites and 8 treatment areas used in this study.

Reclamation site	Area ha	Reclamation year	Time since reclamation	Cover soil type	Number of plots
Site-1	10	2014	2	Forest floor mineral mix	10
Site-2	42	2012	2	Forest floor mineral mix	20
		2012	2	Peat mineral mix	21
Site-3	41	2012	4	Forest floor mineral mix	8
		2012	4	Peat mineral mix	19
Site-4	27	2014	2	Peat mineral mix	18
Site-5	20	2014	2	Forest floor mineral mix	19
		2014	2	Peat mineral mix	4

reclaimed with materials that can facilitate more diverse and productive understory plant communities.

The Athabasca oil sands deposit, located in the boreal forests of northern Alberta, Canada is part of the largest single deposit in the world and covers 142,200 km². Mining of this deposit has transformed a large area of natural boreal forest (nearly 953 km²) into a heavily disturbed industrial landscape (Alberta Environment and Parks, 2017) where vegetation and overburden materials were removed to expose the underlying bitumen laden sands (Fung and Macyk, 2000). After mining, companies in Alberta are required to reclaim the disturbed area according to the Alberta Environmental Protection and Enhancement Act (EPEA) (Alberta Government, 2005). The overarching reclamation goal of EPEA is to achieve locally common boreal forest ecosystems with capability equivalent to pre-development conditions, although individual land uses will not necessarily be identical. While reclaimed lands are expected to function as self-sustaining ecosystems, the common goal of achieving a locally common boreal forest implies that the reclaimed lands will be subject to ecological succession and will change over time. Considering landscape evolution and those successional processes, a balance between ecosystem resiliency and some level of maintenance will be identified and achieved to bring the landscape into a trajectory towards a self-sustaining system.

Generally, reclamation of oil sands mine sites consists of several steps, including cover soil salvage and storage (stockpiling), landform construction, cover soil placement, soil amendment, revegetation, monitoring and certification. Successful reclamation is influenced by activities related to each step and how they work together. The types of material mainly used as a cover soil for oil sands reclamation are peat mineral soil mix (PMM) (ratio of peat to mineral soil mix from 3:2 to 3:4 by volume) and forest floor mineral soil mix (FMM) (upper layer of forest floor materials mixed with underlying mineral soil during salvaging with a ratio of 1:1 to 1:5) (Alberta Environment and Water, 2012). Due to its abundance in the landscape, the peat soil has been historically more commonly used as cover soil (Rowland et al., 2009; Mackenzie and Naeth, 2010; Dhar et al., 2018). Recently FMM has been replacing PMM as an alternate cover soil (Mackenzie and Naeth, 2010; Dhar et al., 2018) due to its greater plant cover, richness and native species abundance over PMM under laboratory (Mackenzie and Naeth, 2010) and field (Errington and Pinno, 2016) conditions. Although these studies provide important insight into the use of different cover soils, further field investigations are required to enhance our understanding of how relative performance of different reclamation strategies can promote diversity and composition of understory plant communities. This understanding is important as understory vegetation represents the majority of forest plant species diversity in boreal forests (Mackenzie and Naeth, 2010; Chen et al., 2017; Dhar et al., 2018) and can be used as an indicator of reclamation success (SER, 2004). Our research objective was to examine whether understory abundance, richness, diversity and community composition differed between cover soil types and time since reclamation. We hypothesized that cover soil type would influence establishment of understory plant community development which is directly linked to reclamation success.

2. Materials and methods

2.1. Study area

Field research was conducted approximately 70 km north of Fort McMurray, Alberta, at Canadian Natural Resources Limited (CNRL) Albion oil sands mine site. The area was within the Central Mixedwood sub-region of the boreal forest with similar climate conditions (Natural Regions Committee, 2006). Mature natural boreal mixed wood forest in this region is dominated by and trembling aspen (*Populus tremuloides* Michx.) and white spruce (*Picea glauca* (Moench) Voss) with components of jack pine (*Pinus banksiana* L.), balsam poplar (*Populus balsamifera* L.), white birch (*Betula papyrifera* Marshall.) and balsam fir (*Abies balsamea* (L.) Mill.) (Natural Regions Committee, 2006). This area has short, warm summers and long, cold winters (Natural Regions Committee, 2006). Mean monthly temperatures are −17.4 °C in January to 17.1 °C in July, with a mean annual temperature of approximately 1.0 °C (Government of Canada, 2017). Total annual precipitation for the area is approximately 418.6 mm, with approximately 55% occurring as rainfall during May through August (Government of Canada, 2017). The majority of soils have developed on glacial and glaciofluvial deposits. Gray Luvisolic soils (based on the Canadian System of Soil Classification (Soil Classification Working Group, 1998) are associated with till and lacustrine deposits, while Dystric Brunisols are associated with glaciofluvial outwash and eolian sands (Beckingham and Archibald, 1996).

2.2. Sampling design and data collection

Five reclamation sites 10–42 ha in size were selected for this study. The sites were reclaimed in 2012 and 2014 using two cover soil types: forest floor mineral mix (FMM) and peat mineral mix (PMM) (Fig. A.1, Table 1). The FMM was from upland forest soil and consisted of surface organic (LFH: an organic horizon containing > 17% organic carbon, approximately ≥30% organic matter by weight) layers mixed with underlying mineral soil (A and B horizons) and salvaged to a maximum depth of 30 cm. The upland forest is dominated by jack pine (*Pinus banksiana*) and aspen (*Populus tremuloides*) and with lesser components of white spruce (*Picea glauca*) and balsam poplar (*Populus balsamifera*). The understory was dominated by *Vaccinium* spp. *Cladonia* spp. and *Cladina* spp. The PMM was from organic soils and consisted of a mixture of peat and underlying mineral soil salvaged at a 60:40 ratio from black spruce (*Picea mariana*) dominated lowland forests. The organic forest floor was 4–5 cm thick. Both PMM and FMM were salvaged and temporarily stockpiled before replacement on the sites and no soil management operations were conducted during storage.

All sites were reconstructed using similar overburden materials (a mixture of soil and organic material salvaged during mine advancement) consisting of lean oil sands (< 10% oil), Pleistocene glacial deposits and Cretaceous silts, shales and sandstones (Macdonald et al., 2012). After landform construction 30 cm of FMM or PMM cover soils were placed over 1.5 m overburden. To minimize compaction both cover soils were uniformly placed across the sites in early spring of the

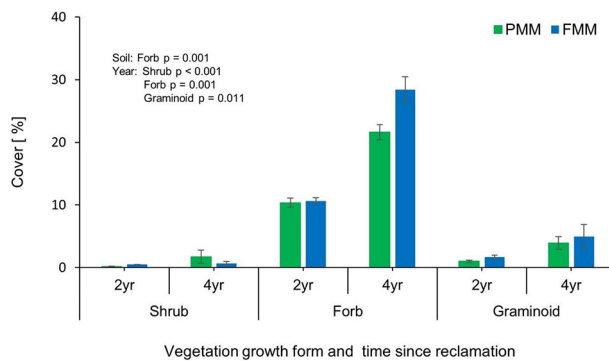


Fig. 1. Percent cover by cover soil type and time since reclamation for three growth forms (shrub, forb, graminoid) in Athabasca oil sands reclamation sites. Bars illustrate mean ($\pm$ SE). (FMM = Forest floor mineral mix, PMM = Peat mineral mix).

reclamation years when the underlying overburden and the PMM layer were still frozen while the surface FMM layer had already thawed.

A grid pattern was used for determination of research plot locations within each reclamation site two and four years after reclamation. The starting point for the grid was randomly chosen using ArcGIS and a uniform 100 m grid point interval was constructed in ArcGIS aligned to the cardinal directions (north, east, south, west) and overlaid onto each reclamation site. Only points located at least 20 m from the boundary of each site were used. Temporary tree stocking plots with 3.99 m radius were established on a 100 m grid. The percent cover of ground vegetation was assessed within 1.5 m radius plots (7.07 m²). The vegetation data were collected two to four years after reclamation and none of the sites was repeatedly measured. Nomenclature for vascular plants followed Moss (1983) as did the description of life strategy or growth form such as trees, shrubs, forbs and graminoids and designation of a species as native and non-native to Alberta. The total number and cover of non-native species and rare species were visually estimated. The total number of sample plots was considered adequate as species numbers reached a plateau for all treatment plots (Fig. A.2, Supporting Information). No soil related data were obtained. After cover soil placement, all reclamation sites were planted with nursery stock recommended by provincial reclamation guidelines (Alberta Environment, 2010). The jack pine, trembling aspen and white spruce were planted at a density of 2500 stems ha⁻¹.

2.3. Statistical analyses

In the statistical analyses, vegetation cover percent and diversity indices including species richness (S), Shannon-Weiner's index (H') and Shannon evenness (E), were used to compare treatments for shrubs, forbs and graminoids.

Linear mixed effects models were used to examine effects of cover soil placement treatments on vegetation cover and diversity indices where cover soil treatment (FMM, PMM) and time since reclamation (2 yr, 4 yr) were treated as fixed effects and site was treated as a random effect. Analyses were conducted using the function lme from the package nlme in the R statistics system version 3.2.5 (R Core Team, 2016). Normality and homogeneity of variance were determined by examining the residuals versus the fitted plots and the normal q-q plots of the models. The square root transformation was carried out when needed.

Nonmetric multidimensional scaling ordination (NMDS) was conducted to examine plant community composition and species assemblages as they related to physical environmental attributes (McCune and Grace, 2002). The ordination was conducted using the package vegan 2.4-2 (Oksanen et al., 2017) in the R statistics system. To support the descriptive ordination and to quantify multivariate differences in

species composition among treatments, a non-parametric multivariate test Multi-Response Permutation Procedure (MRPP) (McCune and Grace, 2002) was conducted. In MRPP analysis the A-statistic describing the chance corrected within group agreement and statistic values over 0.2 indicating a difference between groups (McCune and Grace, 2002). The MRPP analysis was conducted using the function vegdist within the package vegan 2.4-2 (Oksanen et al., 2017). The affinity of species to particular treatment units was evaluated using indicator species analysis on abundance data with the function multi-patt of the package indicspecies version 1.7.5 for the R statistics system (De Cáceres, 2013). Multi-patt is the most commonly used function of indicspecies, allowing determination of species lists associated with particular groups of sites (or combinations) (De Cáceres, 2013).

3. Results

3.1. Cover and diversity

Total percent cover of vegetation groups differed significantly between FMM and PMM cover soil types (forbs $p = 0.001$, graminoids $p < 0.001$) and reclamation year (shrubs $p = 0.022$, forbs $p < 0.001$, graminoids $p = 0.001$). Forbs had the greatest cover (2 yr = 11%, 4 yr = 28%), followed by graminoids (2 yr = 2%, 4 yr = 5%) and shrubs (Fig. 1, Table A.1). The FMM cover was greater than that of PMM for all layers except shrubs 4 years after reclamation (Fig. 1). Native species cover increased with time in both PMM (2 yr = 7, 4 yr = 15%) and FMM (2 yr = 5%, 4 yr = 30%) (Fig. 2a). Cover of non-native species declined with time in FMM and increased in PMM (Fig. 2b). The PMM had greater non-native cover in both years (2 yr = 9, 4 yr = 13%) than the FMM (2 yr = 8, 4 yr = 4%) (Fig. 2b). Two years after reclamation cover of FMM and PMM were not significantly different, although four years after reclamation native cover was significantly greater in FMM and non-native species cover was significantly greater in PMM (Fig. 2a, b).

Species richness and diversity (Shannon and Shannon evenness) differed significantly between treatments and between reclamation years for forbs (richness $p = 0.018$ and $p = 0.015$, Shannon $p = 0.006$ and $p = 0.028$, Shannon evenness $p = 0.041$ and $p = 0.030$). Only species richness and Shannon index differed significantly between FMM and PMM for graminoids (richness $p = 0.019$, Shannon $p = 0.04$). None of the indices showed significant treatment effects on shrub diversity indices except reclamation year for shrub richness ($p = 0.002$ (Table A.1). The FMM had greater richness and diversity (both Shannon index and Shannon evenness) when all layers were combined (Fig. 3). When native and non-native species were compared, species richness and diversity (Shannon and Shannon evenness) differed significantly between FMM and PMM (richness $p = 0.039$, Shannon $p = 0.001$, Shannon evenness $p = 0.012$) and year (richness $p < 0.001$, Shannon $p = 0.026$) except Shannon evenness for reclamation year. Native species richness and diversity were greater in FMM with an increasing trend with time since reclamation (Fig. 2 c, e, g) and a decreasing trend for non-native species (Fig. 2 d, f, h). Richness and diversity of non-native species showed an increasing trend with time in PMM (Fig. 2 d, f, h).

3.2. Community composition

In total 84 understory plant species were found in the reclamation treatments which included 16 shrubs (12 in FMM, 11 in PMM), 48 forbs (41 in FMM, 38 in PMM) and 20 graminoids (18 in FMM, 11 in PMM) (Table A.2). In total 68 native (55 in FMM, 51 in PMM) and 16 non-native (15 in FMM, 10 in PMM) species were identified. Among them native *Epilobium angustifolium* (L.) Scop and *Symphyotrichum boreale* (Torr. & Gray) Nesom and non-native *Melilotus alba* Medik and *Salsola kali* were the most abundant species in FMM and PMM.

Plant community composition did not vary significantly between

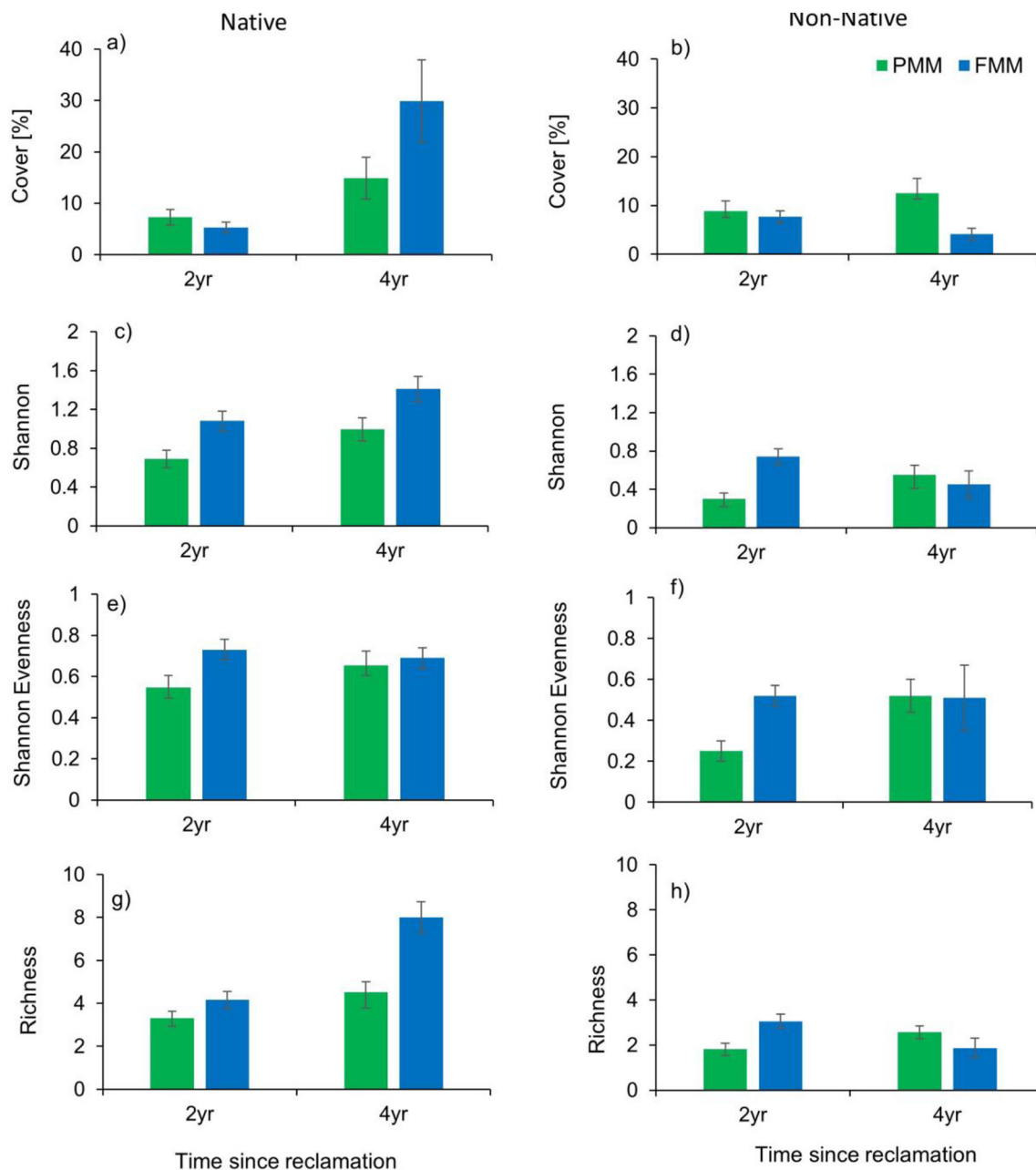


Fig. 2. Percent cover, diversity and richness by cover soil type and reclamation years for native and non-native species in Athabasca oil sands reclamation sites. Bars illustrate mean ($\pm$ SE). (FMM = Forest floor mineral mix, PMM = Peat mineral mix).

FMM and PMM ($A = 0.373$, $T = -1.723$, $p = 0.062$). The non-metric multidimensional scaling (NMDS) ordination did not show strong compositional separation between treatments (Fig. 4), yielded a two dimensional solution, and explained 77.8% of total variation in the original data matrix. The indicator species analysis showed more species were associated with FMM (9 species) than PMM (2 species) (Table 2). In FMM one shrub (*Rubus idaeus* L.), 7 forbs and one graminoid (*Carex aenea*) species were found. Among them five forb species were non-native (*Medicago sativa* L., *Polygonum convolvulus* L., *Chenopodium album* L., *Trifolium pretense* L. and *Trifolium hybridum* L.) and most were perennial.

In PMM two indicator species were found one was non-native annual forb *Salsola kali* L and other was native annual forb *Symphytotrichum ciliatum* (Ledeb.) G.L. Nesom (Table 2). When indicator species analysis was conducted by reclamation years only, two forb species were found two years after reclamation and 14 species four

years after reclamation (Table 2). Another 24 species had the highest indicator value components and were found in both treatments (Table A.2).

4. Discussion

The outcomes of this study support our central hypothesis that cover soil types are the major driver of understory species abundance, diversity and composition following oil sands reclamation. However, soil physical, chemical and biological properties including overstory and understory species interaction may have some influence, which was not considered in this study. The higher understory diversity, richness and vegetation cover on sites reclaimed with FMM were comparable with previous oil sands reclamation studies in the laboratory and field (Mackenzie and Naeth, 2010; Brown and Naeth, 2014; Errington and Pinno, 2016). This may be due to the larger abundance of seeds,

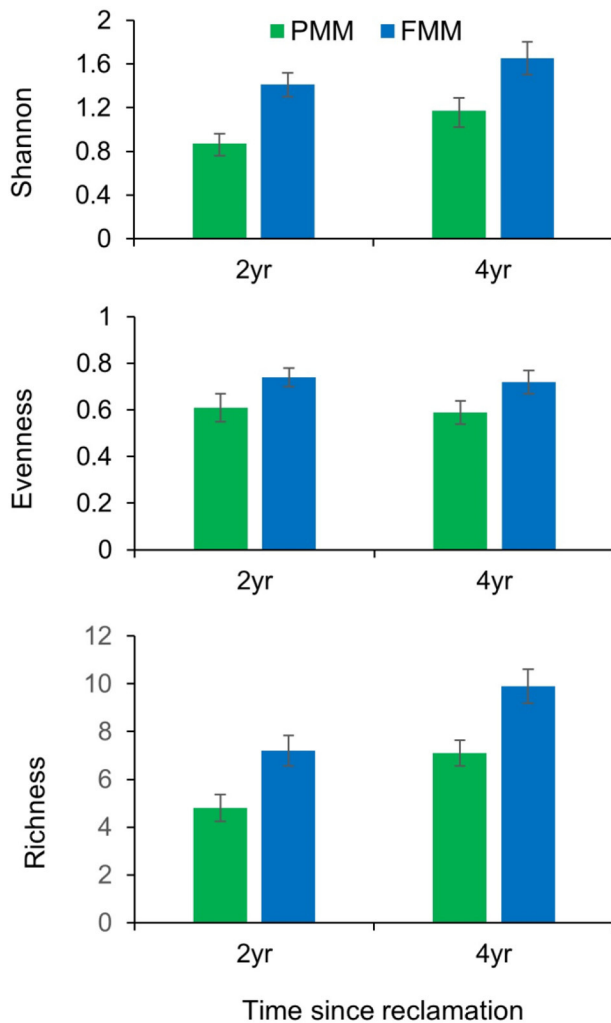


Fig. 3. Species richness and diversity by cover soil type and time since reclamation for three growth forms (shrub, forb, graminoid) in Athabasca oil sands reclamation sites. Bars illustrate mean ($\pm$ SE). (FMM = Forest floor mineral mix, PMM = Peat mineral mix).

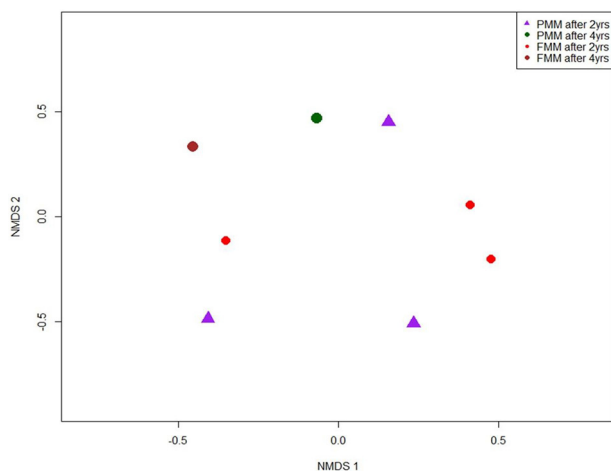


Fig. 4. Nonmetric multidimensional scaling (NMDS) ordination of plant community by cover soil type and time since reclamation in Athabasca oil sands reclamation sites. (FMM = Forest floor mineral mix, PMM = Peat mineral mix).

vegetative propagules and microorganisms present in the LFH horizon of upland forest systems (Mackenzie and Naeth, 2010; Macdonald et al., 2015b; Dhar et al., 2018). The indicator species analysis provided additional support for this conclusion. Since the upland derived cover soil was placed onto an upland reclamation site, the biological legacies (seed and propagules banks, soil ecosystem, microorganisms component) in FMM reclamation sites facilitated remembrance of genetic, compositional and structural condition of natural forests, which could play crucial roles in ecosystem recovery and influence ecosystem processes (Drever et al., 2006; Dhar et al., 2016, 2018, 2020a). The lowest diversity, richness and vegetation cover with PMM was comparable to previous oil sands reclamation studies in this area (Mackenzie and Naeth, 2010; Brown and Naeth, 2014).

Generally, propagules in PMM prefer to grow in subhydric to hydric conditions as peat soil is derived from bogs and fens (Beckingham and Archibald, 1996). Therefore PMM placed on upland sites during reclamation tends to dry, making it difficult for propagules to establish, grow and produce native vegetation cover. Thus reclamation with PMM tends to have cover, diversity and richness of non-native species which are more prominent a few years after reclamation due to their dispersal to the sites. Although the seed bank was not examined in this study, our results are consistent with previous work which indicated that the forest floor and upper soil layers contain more seeds than peat soils (Moore and Wein, 1977; Mackenzie and Naeth, 2010). Reported seed densities in productive boreal forest soils were 1273 to 9108 seeds/m² (Moore and Wein, 1977; Hills and Morris, 1992), whereas Mackenzie and Naeth (2010) found forest floor and peat mineral donor soil propagule banks contained 9000 and 3600 seeds/m², respectively. Of those, 8300 seeds were from native species and 100 from non-native species in the forest floor soil, and 3300 seeds were from native species and 17 from non-native species in peat soil. This may explain why species diversity and richness were higher in FMM than PMM. The dominant cover of forbs and graminoids in the understory and the slow increase in shrub cover were consistent with observations of successional development in the boreal forest. After disturbance boreal ecosystems tend to be dominated by forbs and graminoids, with shrubs becoming a significant component of the ecosystem as succession progresses (Hart and Chen, 2008; Pinno and Hawkes, 2015). The lower rate of increased shrub cover may be due to slower establishment and growth on reclaimed sites, than those of natural disturbances where regeneration occurred from soil seed banks and bud banks (Hart and Chen, 2008; Pinno and Hawkes, 2015). A similar progression of shrub cover has been reported in other studies (Rowland et al., 2009; Pinno and Hawkes, 2015; Dhar et al., 2019, 2020a).

The native species *Rubus idaeus* L., dominant at the FMM reclamation site, is often found in recently disturbed boreal forest communities (Hart and Chen, 2006) and suggests that the successional trajectory after reclamation follows typical boreal forest recovery in FMM. Thus presence of such shrub species is a good indicator of reclamation success. Shrubs accelerate forest floor development, facilitate tree establishment through snow retention, reduced drought and temperature stress and reduced seedling mortality from browsing (Castro et al., 2004; Rowland et al., 2009). The nitrogen fixing shrub *Shepherdia canadensis* (L.) Nutt. (Canada buffalo berry) is present in both FMM and PMM which can increase soil nitrogen pools and soil organic matter (Mummey et al. 2002) and could improve overall soil chemical and biological properties in reclamation sites (Rowland et al., 2009).

Dominance of the perennial community in FMM and annual forbs in PMM show functional groups to which indicator species belong is an important aspect of characterizing plant communities. Trees and shrubs shade ruderal species (Messier et al., 1998; Lieffers et al., 1999), which in turn can reduce competition and promote development of forest understory species. Thus, the trajectory of community development in FMM appears to be consistent with the typical successional progress of boreal forests from ruderal to perennial communities to perennial communities (Landhäusser et al., 1996; Messier et al., 1998; Lieffers

Table 2

Significant ($p < 0.05$) indicator species for the two cover soil types (treatments). [Stat is the association statistic and Pvalue is the p -value of the permutational test. * non-native species, FMM = Forest floor mineral mix, PMM = Peat mineral mix].

Treatment	Species	Stat	P value	Common Name	Type	Life strategy
<i>By cover soil treatment</i>						
FMM	<i>Rubus idaeus</i>	0.341	0.048	Wild red raspberry	Shrub	Perennial
	<i>Medicago sativa</i> *	0.519	0.012	Alfalfa	Forb	Perennial
	<i>Polygonum convolvulus</i> *	0.478	0.001	Wild buckwheat	Forb	Annual
	<i>Geranium bicknellii</i>	0.463	0.019	Bicknell's geranium	Forb	Annual / Biennial
	<i>Chenopodium album</i> *	0.542	0.003	Lamb's quarters	Forb	Annual
	<i>Polygonum lapathifolium</i>	0.530	0.001	Pale persicaria	Forb	Annual
	<i>Trifolium pratense</i> *	0.530	0.001	Red clover	Forb	Biennial / Perennial
	<i>Trifolium hybridum</i> *	0.463	0.009	Alsike clover	Forb	Perennial
	<i>Carex aenea</i>	0.334	0.018	Silvery flowered sedge	Graminoid	Perennial
PMM	<i>Salsola kali</i> *	0.535	0.001	Russian thistle	Forb	Annual
	<i>Symphyotrichum ciliatum</i>	0.492	0.001	Rayless aster	Forb	Annual
<i>By time since reclamation</i>						
After two years	<i>Trifolium pratense</i> *	0.454	0.027	Red clover	Forb	Biennial / Perennial
	<i>Polygonum lapathifolium</i>	0.430	0.034	Pale persicaria	Forb	Annual
After 4 years	<i>Arctostaphylos uva-ursi</i>	0.333	0.011	Common bearberry	Shrub	Perennial
	<i>Sonchus arvensis</i> L	0.823	0.001	Perennial sow thistle	Forb	Perennial
	<i>Ribes oxycanthoides</i>	0.333	0.015	Northern gooseberry	Shrub	Perennial
	<i>Epilobium angustifolium</i>	0.747	0.001	Common fireweed	Forb	Perennial
	<i>Symphyotrichum ciliatum</i>	0.719	0.001	Rayless aster	Forb	Annual
	<i>Potentilla norvegica</i>	0.718	0.006	Rough cinquefoil	Forb	Annual / Biennial
	<i>Hieracium umbellatum</i>	0.675	0.007	Narrow leaved hawkweed	Forb	Perennial
	<i>Salsola kali</i> *	0.580	0.012	Russian thistle	Forb	Annual
	<i>Symphyotrichum ciliatum</i>	0.577	0.001	Lindley's aster	Forb	Annual
	<i>Fragaria virginiana</i>	0.577	0.001	Wild strawberry	Forb	Perennial
	<i>Equisetum arvense</i>	0.0327	0.046	Common horsetail	Forb	Perennial
	<i>Tragopogon dubius</i>	0.272	0.044	Common goat's beard	Forb	Annual / Biennial
	<i>Hordeum jubatum</i>	0.629	0.001	Foxtail barley	Graminoid	Perennial
	<i>Calamagrostis canadensis</i>	0.380	0.013	Bluejoint	Graminoid	Perennial

et al., 1999; Hart and Chen, 2006; Dhar et al., 2018). Dominance by annual species in the PMM could slow successional progress and ecosystem recovery. Increasing indicator species with time indicates propagule dispersal plays a significant role in plant community development. The most important factor is that only one non-native and few annual indicator species were found after four years, suggesting that with time non-native species presence may decline and plant community development could progress towards that typical of natural boreal forest ecosystems.

Non-native species can cause a loss of biological diversity and can be problematic in reclaimed areas, although in some cases they can reduce soil erosion and contribute to soil development (Dhar et al., 2018). After four growing seasons the diversity, richness and cover of non-native species were greater on PMM than FMM, which was supported by a previous study that found greater non-native species cover on PMM five years after reclamation (Forsch, 2014). Higher non-native species in PMM might be linked to organic carbon in the cover soil types. Forsch (2014) found greater non-native species cover on PMM (6.62%) where organic carbon content was low relative to FMM (7.34%). Lower organic carbon generally implies a greater mineral component of the seedbed and facilitates seedling establishment (Nyland, 1996). Some studies found that both number and cover of non-native species were greater on FMM than PMM (Mackenzie and Naeth, 2010; Errington and Pinno, 2016). However, a declining trend with time in FMM (7.5–4%) indicates that non-native species in FMM might be exploiting the resources abundantly available after land reclamation and will be eliminated over time as native species develop. An increasing trend for non-native species cover in PMM (8.9–12.5%) at the initial stage of reclamation is comparable to other long term studies (Pinno and Hawkes, 2015; Dhar et al., 2019, 2020a). Pinno and Hawkes (2015) found non-native species cover in PMM increased up to 5 years after reclamation and declined thereafter. Similarly, Dhar et al. (2020a) found non-native species cover declined from 32 to 10% within

24 years after reclamation. Dhar et al. (2019) reported that PMM had less than 5% non-native species cover 19 to 24 years after reclamation. While these studies indicate that non-native species in PMM will decline with time, ongoing research with continuous monitoring is needed to document the potential long term impacts of non-native species on ecosystem recovery of reclaimed sites if PMM is used as a cover soil.

While FMM provide better environment for plant community development than PMM, it is not always possible to use FMM cover soil due its scarcity on the landscape. Therefore, some manipulation and management activity may be required to enhance plant community development if PMM is used as a cover soil. Such measures could include mixing peat with forest floor soil during cover soil placement. If mixing is difficult, FMM cover soil could be randomly or systematically placed in small patches (bands, piles) on top of PMM. The propagules from these piles could disperse into surrounding reclamation areas with time.

5. Conclusions

This study provided some insights relating to the use of PMM and FMM as a cover soil in oil sands reclamation where use of FMM can provide better reclamation success for plant community development, vegetation cover, diversity, richness and controlling non-native species. Although compositional differences between the cover soil types were not evident, FMM cover soil was dominated by perennial species, while PMM was dominated by annual forb species. The trajectory of plant community development in the FMM cover soil led to faster recovery (from ruderal and annual communities to perennial communities) than PMM cover soil, which was the typical early successional progression of boreal forests in this region. Continued monitoring is required to quantify ecosystem recovery following reclamation of oil sands mines. Several parameters that could be used to characterize the cover soils such as water holding capacity, soil water content, soil temperature,

soil chemical properties (available nutrients and soluble ions) and properties of the soil microbial community were not estimated. In future monitoring studies information on these factors could contribute to improvements in reclamation practise.

Declaration of Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Competing Interest

None.

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Appendix A. Supplementary data

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